



Supplementary Materials for

China's solar expansion policy reduces bird diversity

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Science **393**, 831 (2026)
DOI: 10.1126/science.aee0747

The PDF file includes:

Materials and Methods
Supplementary Text
Figs. S1 to S4
Tables S1 to S19
References

Other Supplementary Material for this manuscript includes the following:

MDAR Reproducibility Checklist

Materials and Methods-Data description

This study was based on a novel, large-scale panel dataset constructed to cover 2,344 counties across China for the period from 2014 to 2023, yielding a total of 46,528 county-month observations. The dataset integrated information from multiple sources to create a comprehensive picture of policy, biodiversity, and environmental conditions (Table S2).

Our bird observation data were sourced from the China Birdreport Online Database, a large-scale citizen science dataset curated by the China Birdwatching Record Center (www.birdreport.cn). This platform aggregated user-submitted records of bird sightings, each containing precise spatiotemporal coordinates and species-specific abundance counts (22, 24). The official conservation status for species in China was derived from *the List of State Key Protected Wild Animals (2021)*, which was jointly promulgated by the National Forestry and Grassland Administration and the Ministry of Agriculture and Rural Affairs. The richness and geographic breadth of this dataset allowed for a fine-grained analysis of bird diversity dynamics across the vast and heterogeneous landscape of China.

The central explanatory variable, the photovoltaic policy stringency index (PSI), was constructed through a data collection and text analysis process. We systematically scraped the official government websites of each county and its respective prefecture-level city, as well as national-level portals, to collect all official documents related to the promotion of solar PV development. The collected policies were then scored based on their level of mandatory enforcement (Table S3). To operationalize this scoring framework, our approach first distinguished between nine official formats of Chinese photovoltaic power generation policies, ranking them on a scale of 1 to 9 based on their administrative binding force. The scale begins with Outlines and Plans, which offer broad guidance and strategic vision, followed by Key Points and Opinions that provide specific priorities and advisory guidelines. Mid-tier documents include Implementation Schemes and Notices, which focus on concrete execution plans and the transmission of requirements. The hierarchy culminates in Administrative Measures, Regulations, and Ordinances, which represent the strongest binding force through codified rules and quasi-legislative authority. Furthermore, we applied a differential weighting scheme to administrative levels, drawing on theories of the pressure-based system and administrative subcontracting in Chinese governance. Provincial governments receive the highest weight as the critical nexus between central and local administration. They wield substantial power over resource allocation and personnel, and their policies often carry severe one-vote veto assessment pressures. Municipal governments, functioning as intermediaries that decompose targets and supervise implementation, are assigned a moderate weight. Conversely, county governments are weighted lowest, while they serve as the ultimate implementers with some discretionary execution power, their policy documents typically function as operational details or mobilization orders rather than binding mandates. This multi-level approach could reflect the nested structure of policy implementation in China.

For validating our policy stringency index functions as a proxy for concrete development outcomes, we correlated it with county-level data on land area of centralized solar photovoltaic (PV) plants. The analysis uncovered a positive and statistically significant correlation. This consistent trend was further corroborated by a binned scatter plot showing a significant positive association between PV policy stringency and PV area ($\beta = 0.0222$, $SE = 0.0026$, $p < 0.01$; Fig. S2). Our findings remained robust to a sensitivity test, when remeasured the policy stringency

using alternative scoring scales.

To isolate the effect of policy, we incorporated a comprehensive suite of control variables to account for potential confounding factors. Climatic variables, including annual average temperature (Temp) and wind speed (Wind), were obtained from the ERA5-Land reanalysis dataset, a state-of-the-art global climate dataset. Socio-economic and observational controls include population density (Pop), sourced from the National Bureau of Statistics; birdwatching effect, proxied by birdwatching time per capita (Duration) from the China Birdwatching Report, and carbon emissions (Carbon), derived from the Emissions Database for Global Atmospheric Research (EDGAR). Finally, land cover characteristics, critical for bird habitats, were measured as the proportional area of water bodies (Water), forests and shrubs (Green), farmland (Farm), and grassland (Grass). These metrics were calculated based on the 30-meter resolution China land cover dataset. The annual agricultural crop yield data was sourced from the China County Economy Research Database, which is hosted within the China Stock Market & Accounting Research (CSMAR) system (<https://data.csmar.com/>). The data is reported on an annual basis. The original unit of measurement for the crop yield is tons. To construct the panel dataset across the 2,344 counties, we first matched and aggregated the annual yield data using standardized county-level administrative codes. Subsequently, we applied a natural logarithm transformation to the total yield values.

Materials and Methods-Descriptive statistics

Table S4 presents the descriptive statistics for the main variables used in our analysis, covering the full sample from 2014 to 2023. The dataset contains 46,528 county-month observations for most variables. The dependent variable, the Shannon Index of bird diversity, has a mean of 2.446, with a standard deviation of 0.981, indicating a moderate level of diversity on average, but with substantial variation across the sample. Our primary independent variable, photovoltaic policy stringency index (PSI), has a mean of 2.662. Critically for our identification strategy, it also exhibits a large standard deviation (4.116) and ranges from 0 to 15.5, which confirms that there is significant variation in the stringency of policy implementation across counties and over time. This variation is essential for the econometric analysis to identify the policy's effect. Descriptive statistics for the key heterogeneity groups are detailed in Table S5. Fig. 1 and fig. S1 delineate the spatial patterns and temporal evolution of the PV policy stringency index, power station deployment, and bird diversity between 2014 and 2023, while also displaying the distribution of these variables across counties via violin plots.

The control variables reflect the wide range of environmental and demographic conditions across the sample area. The annual average temperature (Temp) is 14.83°C, and the annual average wind speed (Wind) is 2.24 m/s. Natural environmental factors, which are critical determinants of bird habitats, also show considerable heterogeneity. Specifically, farmland (Farm) and the combined coverage of forests and shrubs (Green) dominate the landscape, accounting for 36.34% and 33.18% on average, respectively. In contrast, the proportions of grassland (Grass) and water bodies (Water) are relatively small, at 8.93% and 3.58%. However, despite their smaller overall shares, they demonstrate notable variation across the regions and over time, with standard deviations of 0.21 for Grass and 0.05 for Water. The population density (Pop), a proxy for human activity, has a mean of 5.57 in its logarithmic form. The logarithm of carbon emissions (Carbon) has a mean of 14.56, and the logarithm of birdwatching duration per birdwatcher (Duration) averages 1.50. The substantial variation observed across all variables provides a robust foundation for the subsequent regression analyses, allowing us to control for a diverse set of local conditions while estimating the central relationship of interest.

Materials and Methods-Method details

We discuss how the rapid, policy-driven expansion of photovoltaic (PV) solar power, a cornerstone of global climate mitigation strategies, could create a complex “green-green dilemma” by pitting decarbonization objectives against biodiversity conservation goals. The ecological consequences of this energy transition, particularly for bird diversity, remained contentious and theoretically ambiguous. The literature presented two competing causal pathways through which intensive PV development policies may influence bird diversity. One narrative posited that such policies can yield a net positive for avian communities, primarily by catalyzing a shift away from fossil fuels, thereby alleviating the broader environmental pressures of climate change and air pollution, which are themselves primary drivers of biodiversity loss (5). Climate change, for instance, disrupts fundamental ecological functions and has been shown to directly reduce animal diversity through mechanisms such as the intensification of El Niño events (35), while atmospheric pollutants are known to negatively impact bird populations (36).

On a more localized scale, this ecological enhancement pathway suggests that solar parks can create novel habitats that are more beneficial to birds than the land uses they replace, a scenario particularly relevant in highly modified agricultural landscapes (25). Industrial agriculture is a well-documented driver of terrestrial biodiversity loss, a result of habitat homogenization, widespread pesticide use, and intensive management practices that simplify ecosystems (37–40). In stark contrast, solar parks often feature significantly lower management intensity and can increase habitat heterogeneity at multiple spatial scales (41–43). The physical structures of PV arrays can provide perching and nesting sites for certain species and offer refuge from avian predators, functioning as “habitat islands” or refuges in otherwise degraded landscapes (12, 13). In addition, policy can mandate or incentivize management practices that amplify these benefits. For example, requirements to plant native vegetation within solar park perimeters can increase local plant diversity (10), which in turn supports larger and more diverse invertebrate populations, a critical food source for many bird guilds, especially insectivores (44–46). The unique microclimates created beneath solar panels, characterized by increased local humidity and greater thermal heterogeneity, can also contribute to biodiversity gains (47). By steering development toward marginal or low-conservation-value lands, policy can further minimize conflicts with pristine ecosystems (48, 49). Through these interconnected mechanisms, displacing more harmful energy sources, creating higher-quality habitat than intensive agriculture, and promoting beneficial on-site management, a strong, ecologically-informed PV policy framework could foster positive outcomes for bird communities. This leads to our first hypothesis:

H1: The stringency of photovoltaic policies will positively impact bird diversity.

Conversely, a substantial body of evidence supported an opposing “habitat degradation” pathway, wherein a rapid, policy-driven expansion of solar energy negatively impacts bird diversity. The central mechanism for this adverse impact is the extensive land requirement of solar PV technology, which has a significantly lower power density compared to conventional fossil fuel energy sources (50, 51). Aggressive PV policies, designed to meet ambitious renewable energy targets, can consequently drive massive land-use change, leading to the direct

loss, fragmentation, and degradation of natural and semi-natural habitats (18, 52, 53). This land-use conflict is a primary and well-established threat to global biodiversity (54–56). Research demonstrated that renewable energy development, including solar, posed a threat to many globally important biodiversity areas (9), and that there existed a significant spatial overlap between regions with high solar generation potential and the ranges of threatened bird species (21, 57). The construction and operation of large-scale solar facilities transformed these landscapes into industrial zones, destroying critical foraging, breeding, and nesting grounds for native avifauna (27, 58). This process of habitat loss is a well-understood cause of population decline and heightened extinction risk for numerous species worldwide (27, 28).

Beyond direct habitat destruction, solar facilities introduce a suite of novel ecological risks. Direct mortality can occur through the “lake effect”, a phenomenon where birds, particularly water-dependent species, apparently mistake the reflective surfaces of PV panels for water bodies and collide with them, often leading to severe injury or death (59, 60). Furthermore, the ecological disturbance from this infrastructure often extends far beyond its physical footprint, creating a zone of influence where birds may avoid the surrounding area due to heightened human activity, noise, and the altered visual landscape (61). This avoidance behavior effectively results in a functional loss of habitat that is much larger than the project area itself. From a policy perspective, when the primary objective is the rapid deployment of energy production capacity, ecological considerations can become secondary (62, 63). When photovoltaic (PV) policies are not reinforced by legal provisions such as mandatory strategic environmental assessments and clear, science-based siting criteria, they inevitably lead to the prioritization of energy objectives over the preservation of biodiversity (19, 20). The intense political and economic push to meet climate goals can thus result in severe, unintended consequences for wildlife (64, 65), suggesting that the stronger the policy incentive for PV expansion, the greater the pressure on land and the more significant the negative impact on bird populations. This leads to our second, opposing hypothesis:

H2: The stringency of photovoltaic policies will negatively impact bird diversity.

Bird Diversity Index

To quantify the complexity and health of local bird communities, we employed the Shannon Index, a cornerstone metric in ecology that measures biodiversity by integrating both species richness (the number of distinct species) and species evenness (the relative abundance of individuals among those species). Its use is motivated by the ecological principle that a diverse and resilient ecosystem is not only rich in species but also maintains a balanced community structure, free from the overwhelming dominance of a few generalist species (25). A higher Shannon Index value is indicative of a healthier, more complex, and more stable ecosystem. By calculating this index for each county-month observation, we capture a dynamic and nuanced picture of local ecological conditions over time. The Shannon Index is calculated as the negative sum of the proportion of each bird species multiplied by its natural logarithm:

$$ShannonBD = -\sum_{i=1}^S x_i \ln x_i, \quad (1)$$

where x_i represents the relative abundance of the i -th bird species in the community.

For our robustness checks, we employed an alternative biodiversity metric, the Simpson

Diversity Index. While the Shannon Index is more sensitive to the presence of rare species, the Simpson Index places greater weight on the most abundant species in a community, thus providing a different perspective on community structure. Its use allows us to confirm that our findings are not merely an artifact of the specific mathematical properties of one index, but hold across different conceptualizations of biodiversity. The Simpson Diversity Index is calculated as:

$$SimpsonBD = 1 - \frac{\sum_{i=1}^s n_i^2}{N^2}, \quad (2)$$

where n_i is the number of observed individuals of the i -th bird species, and N is the total number of individuals of all species in the community.

Baseline Regression Model

To estimate the impact of photovoltaic policy stringency on bird diversity, we employed a high-dimensional fixed-effects regression model. This quasi-experimental approach is designed to isolate the causal effect of policy from a wide range of potential confounding factors. Our baseline specification is as follows:

$$ShannonBD_{cym} = \alpha + \beta PSI_{cym} + \gamma Z_{cym} + \lambda_c + \mu_{ym} + \varepsilon_{cym}. \quad (3)$$

The dependent variable, $ShannonBD_{cym}$ represents bird diversity for county c during month m of year y , as measured by the Shannon Index. The key independent variable PSI_{cym} denotes the stringency of PV power generation policy, a novel index we construct to capture the stringency and cumulative weight of governmental support for solar development. The model includes a vector of time-varying control variables, Z_{cym} that account for confounding geographic, climatic, and anthropogenic factors known to influence bird habitats. These include annual average temperature, annual average wind speed, population density, birdwatching duration, carbon emissions, and the proportional coverage of water bodies, forests and shrubs and farmland. Main variable definitions are detailed in Table S1.

The model incorporates a two-way fixed-effects (TWFE) structure to mitigate omitted variable bias, a pervasive challenge in non-experimental policy evaluation. The inclusion of county fixed effects (λ_c) absorbs all time-invariant unobserved heterogeneity at the county level. This specification controls for stable, county-specific attributes, such as baseline ecological conditions, fundamental geographic characteristics, and local cultural attitudes toward conservation that could otherwise confound the relationship between policy and biodiversity. Complementing these spatial controls, year-month fixed effects (μ_{ym}) account for common temporal shocks, trends, and seasonality that affect all counties at the aggregate level. This approach neutralizes the influence of factors like nationwide macroeconomic trends, large-scale climate patterns (e.g., El Niño years), and aggregate seasonal migration patterns that could systematically influence bird observations across the country. This TWFE design allows for a more credible estimation of the policy impact by ensuring that the coefficient β is identified from within-county changes in policy stringency over time, after netting out common time trends. To account for potential serial correlation in the error term ε_{cym} within counties over time, standard errors are clustered at the county level.

Supplementary text-Robustness checks

To validate a causal interpretation of our baseline results, we subjected our findings to a series of robustness tests. These tests were designed to systematically address potential econometric challenges that could confound our analysis, including endogeneity from omitted variables or reverse causality, non-random selection into policy adoption, sensitivity to variable measurement, the influence of confounding policies, and potential spatial or seasonal variations. By addressing these issues, we confirmed the reliability and stability of our core finding that increased photovoltaic policy stringency is associated with reduced bird diversity. Specifically, we presented results from seven distinct approaches: (i) instrumental variable regressions to address endogeneity; (ii) propensity score matching to account for non-random policy assignment; (iii) the use of alternative variable measures to ensure results are not dependent on a single metric; (iv) sensitivity tests using alternative sampling criteria to check for the influence of outliers and specific administrative regions; (v) exclusion of county-month observations below minimum birdwatching record thresholds to address potential measurement error from limited observer effort; (vi) the inclusion of multiple confounding policy controls; and (vii) additional robustness checks that control for air pollution ($PM_{2.5}$), address potential spatial spillovers by excluding adjacent counties, and verify seasonal stability during the peak breeding season.

Robustness check-Instrumental variable regression

A central concern in evaluating the ecological impacts of policy is endogeneity, which can arise from two primary sources. First, omitted variable bias may occur if unobserved local characteristics, such as a government's underlying environmental preferences or the intensity of local conservation activism, simultaneously influence both the enactment of strong PV policies and local biodiversity outcomes. Second, reverse causality could confound estimates; for instance, regions with already degraded ecosystems and low biodiversity might be preferentially targeted for large-scale PV projects to minimize environmental conflicts, creating a spurious correlation.

To address these endogeneity concerns and isolate the causal effect of photovoltaic policy stringency index (PSI) on bird diversity (ShannonBD), we employed a two-stage least squares (2SLS) instrumental variable (IV) approach. We constructed a new instrument: $\text{Ln}(\text{Sunshine}) / \text{city CPU}$, defined as the natural logarithm of the historical average sunshine duration over a 30-year period (1984-2013) divided by the city-level climate policy uncertainty (CPU) index. The CPU index is a news-based metric constructed by Ma et al. (66), which follows the foundational text-mining methodology of Baker et al. (67) to capture the unpredictability of climate-related policy actions. Structurally, this instrument operates within a shift-share (Bartik) framework, where historical sunshine acts as the time-invariant geographic "share" and the inverse of the news-based CPU index serves as the time-varying "shift." This ratio formulation captures the directional drivers of PV adoption, scaling positively with historical solar endowment (which dictates technological potential) and inversely with climate policy uncertainty (which deters infrastructure commitments). This instrument satisfies the relevance condition because a region's historical solar endowment, combined with macro-level climate policy uncertainty, predicts the likelihood and scale of local PV policy implementation. Furthermore, the exclusion restriction is plausible: historical weather patterns from decades

past and macro-level city policy uncertainties are predetermined and exogenous to contemporary, county-level ecological dynamics, affecting local bird diversity only through their influence on PV facility siting and policy promotion.

Table S7 presents the 2SLS regression results. Column (1) reports the first-stage regression, demonstrating that the instrument is relevant. The coefficient for $\text{Ln}(\text{Sunshine}) / \text{city CPU}$ is 0.0252 and is significant ($\text{SE} = 0.0046$, $p < 0.01$). The validity of this instrument is further confirmed by statistical testing: the Kleibergen-Paap (K-P) rk LM statistic is 30.780, which rejects the null hypothesis of underidentification. The K-P rk Wald F statistic is 30.181 (with a Cragg-Donald Wald F statistic of 664.359), which exceeds the conventional threshold of 10, thereby mitigating any concerns regarding weak instrument bias (68).

Column (2) reports the second-stage results. After isolating the exogenous variation in policy stringency, the estimated coefficient on PSI is -0.1264 ($\text{SE} = 0.0347$, $p < 0.01$). This robust IV estimate not only confirms the negative direction of the relationship but also reveals that the ecological cost of PV expansion on local avian communities is significant and robust, even after accounting for potential endogeneity such as the strategic siting of PV projects in areas with already lower-than-average biodiversity.

To validate the shift-share architecture of our instrument, we conducted an extensive suite of diagnostics. Following the quasi-experimental design formalized by Borusyak et al. (69), we verified the exogeneity of the shift component; after absorbing city and year fixed effects, local ecological and socioeconomic covariates do not systematically predict within-city temporal variation in the CPU index (Table S8). Furthermore, applying the diagnostic framework recommended by Goldsmith-Pinkham et al. (70), we constructed multiple instruments by splitting the shift component into terciles based on the CPU level and interacting each tercile indicator with the original instrument, yielding three instruments that exploit the same within-city temporal fluctuation in climate policy uncertainty but operate under different policy uncertainty regimes (low, medium, and high CPU). As shown in Table S9, the Hansen J statistic is 0.6356, failing to reject the null hypothesis that all instruments are jointly valid at conventional significance levels. Moreover, the estimated coefficient on PSI in the overidentified model remains negative, confirming that the instrument identifies a stable negative effect of photovoltaic policy stringency on bird diversity regardless of whether the shift operates under low, medium, or high policy uncertainty conditions. Sensitivity analyses also demonstrated that our estimates remain stable when sequentially excluding regions with extreme top and bottom percentiles of historical sunshine, confirming our results are not driven by high-leverage outliers. Finally, we conducted a series of shift-based IV identification tests following Goldsmith-Pinkham et al. (70) and Borusyak et al. (69) to further validate our instrument (Table S10, Columns 1–5). Specifically, the balance test reported in Column (1) reveals that although $\text{Ln}(\text{Sunshine})$ is correlated with certain covariates, such cross-sectional variation is entirely absorbed by the county fixed effects in our main specification. To further ensure that sunshine does not directly influence bird diversity through alternative temporal channels, Columns (2)–(3) additionally control for $\text{Ln}(\text{Sunshine}) \times \text{Year-month}$ fixed effects, thereby isolating the identification to within-county policy fluctuations. In Columns (4)–(5), we go a step further by employing a pure shift instrument ($1/\text{CPU}$) that omits the sunshine share component entirely. The consistently negative and statistically significant estimates across these stringent specifications confirm that our findings are driven by the exogenous rollout of

photovoltaic policies rather than by endogenous baseline characteristics, thereby cementing the validity of the exclusion restriction.

A binscatter plot showing the partial correlation between the instrument (the natural logarithm of average sunshine duration from 1984 to 2013 divided by the city-level climate policy uncertainty index) and the endogenous variable (PSI) was included in Fig. S3. The positive slope ($\beta = 0.0401$, $SE = 0.0028$, $p < 0.01$) confirms that regions with greater sunshine endowment and less local climate policy uncertainty tend to exhibit stronger photovoltaic policy stringency, supporting the relevance condition required for a valid instrument.

Robustness check-Propensity score matching

Another potential issue is selection bias stemming from the non-random assignment of PV policies. Counties implementing intensive PV policies (the "treated" group) may systematically differ from those that do not (the "control" group) along observable dimensions, such as economic structure, population density, or pre-existing land-use patterns. If these differences correlate with biodiversity outcomes, a simple comparison will yield biased estimates. Propensity score matching (PSM) mitigates this by constructing a more comparable control group (71).

To ensure the robustness of our results, we applied three matching algorithms: 1:1 nearest neighbor matching (caliper = 0.05), 1:2 nearest neighbor matching (caliper = 0.05), and kernel matching. The nearest neighbor approach pairs each treated unit with the closest control unit(s) within a specified propensity score radius, forming strict matches between comparable observations. In contrast, kernel matching constructs the counterfactual outcome using a weighted average of all control units, thereby incorporating more information from the control group. The large difference in the number of observations between the two methods is due to their distinct matching algorithms. Nearest neighbor matching employs strict 1:1 (or 1:2) pairing within a common support, dropping distant unmatched samples to maximize precision. Conversely, kernel matching retains a larger sample size by using a weighted average of all available control group samples. The PSM results (Table S11) estimate the average treatment effect on the treated (ATT). Column (1), using 1:1 nearest neighbor matching, yields a PSI coefficient of -0.0072 ($SE = 0.0039$, $p < 0.10$). Increasing the matching ratio to 1:2 (Column 2) produces a coefficient of -0.0083 ($SE = 0.0038$, $p < 0.05$). Finally, the kernel matching method (Column 3) results in a significant coefficient of -0.0125 ($SE = 0.0037$, $p < 0.01$).

The consistency of these negative estimates across different matching specifications reinforces our baseline findings. By establishing a statistically comparable control group, PSM effectively isolates the effect of PV policies from confounding regional characteristics. The persistent, statistically significant negative coefficients confirm that the observed adverse impact on bird diversity is not an artifact of systematic differences between regions, providing robust evidence that the implementation of intensive PV policies causally diminishes local bird diversity.

Robustness check-Alternative variable definitions

The robustness of our findings could be contingent on the specific ways we have defined and measured our core variables: the dependent variable (bird diversity) and the key independent variable (photovoltaic policy stringency index). Measurement error or reliance on a single, potentially narrow indicator for complex socio-ecological concepts can lead to misleading conclusions. To address this, we re-estimated our baseline model using a comprehensive suite of alternative measures for both photovoltaic policy stringency and bird diversity.

First, we substituted our primary independent variable, PSI (a weighted policy index), with alternative conceptualizations of policy impact and physical footprint. In Column (1) of Table S12, we replaced PSI with the total area of centralized PV power stations within a county. This measure captures the physical footprint of PV development rather than the policy signal, providing a direct proxy for land-use conversion. We calculated the Pearson correlation coefficient between the policy stringency index (PSI) and the area of centralized photovoltaic power plants (Area), revealing a modest but statistically significant positive correlation (*Pearson's* $r = 0.041$, $p < 0.01$). Importantly, this modest contemporaneous correlation is expected given the fact that large-scale PV development is affected by many other factors such as solar potential and land availability. Operating at a granular county-month level, our data captures the inherent time lag between immediate policy announcements and delayed physical construction (Table S6). To elucidate this relationship, we constructed a binned scatterplot of PSI against Area (Fig. S2). Specifically, the x-axis variable (Area) was divided into equal-sized bins, and the mean of the y-axis variable (PSI) within each bin was plotted. The plot exhibits a positive and upward-sloping trend line ($\beta = 0.0222$, $SE = 0.0026$, $p < 0.01$), indicating that, on average, an increase in policy stringency is associated with an expansion in the actual physical footprint of centralized photovoltaic power plants.

We further updated the time-series distributions of PSI and Area presented in Fig. S1. During the early periods, the distributions of both PSI and Area exhibited right skewness. The medians were near zero, yet the upper tails were exceptionally long and punctuated by sporadic high-value outliers, reflecting a period characterized by localized photovoltaic development. Following China's formal announcement of the "30-60" carbon peaking and carbon neutrality goals in late 2020, the time-series data reveal a massive, synchronous upward trend. In both panels A and B, the violin plots for 2023 display substantial expansion: the interquartile ranges extend upward, the density of high-value observations surges, and the medians begin to ascend. This visually captures the nationwide policy boom, as the policy focus shifted from poverty alleviation to the large-scale, ubiquitous deployment of utility-scale photovoltaic power. This synchronous temporal evolution further corroborates that our policy stringency variable effectively captures the scale expansion of solar infrastructure over time.

As another check on the independent variable, we conducted robustness tests by re-categorizing the nine distinct policy types into three broader functional classes to reflect varying levels of coerciveness: Strategic Planning (comprising Outlines, Plans, and Key Points), Executive Guidance (including Opinions, Implementation Schemes, and Notices), and Normative Management (consisting of Administrative Measures, Regulations, and Ordinances). We then re-measured policy stringency using three alternative scoring progressions to test for sensitivity: an arithmetic progression of 1, 2, and 3 points; a geometric progression of 1, 2, and 4 points; and an exponential progression of 2, 4, and 8 points. These alternative scoring systems

were applied using both our original hierarchical weights (Columns 2-4) and equal weights (Columns 5-7).

Second, we replaced our dependent variable, the Shannon Index (ShannonBD), with the Simpson Diversity Index (SimpsonBD) in Column (8). While both are established biodiversity metrics, the Shannon Index is more sensitive to species richness, whereas the Simpson Diversity Index gives greater weight to species evenness (the relative abundance of species). A robust ecological finding should hold across different facets of biodiversity.

Table S12 presents the results from these alternative specifications. In Column (1), regressing the Shannon Index on the physical PV area yields a negative and significant coefficient ($\beta = -0.0387$, $SE = 0.0118$, $p < 0.01$), confirming that a larger physical PV footprint is directly associated with lower bird diversity. This provided the evidence that large, continuous solar fields plausibly have stronger impacts than smaller or more fragmented projects. In Columns (2) through (7), using the re-categorized policy stringency measures across all arithmetic, geometric, and exponential scoring progressions, under both hierarchical and equal weighting schemes, the coefficients remain negative and statistically significant (β ranging from -0.0325 to -0.0061 , SE ranging from 0.0090 to 0.0017 , $p < 0.01$ for all).

Finally, in Column (8), utilizing the Simpson Diversity Index as the dependent variable with our original PSI measure, we again observe a significant negative effect ($\beta = -0.0014$, $SE = 0.0008$, $p < 0.10$). The negative relationship between PV promotion and bird diversity is robust to varied conceptualizations and measurements of both the policy intervention and the ecological outcome, confirming the validity of our primary findings.

Robustness check-Excluding the influence of outliers and specific regions

The overall average effect estimated in our baseline model could be disproportionately influenced by a small number of observations with extreme values (outliers) or by specific types of regions with unique characteristics. For example, large-scale PV projects in a few counties could drive the results, or the dynamics in major metropolitan areas might differ fundamentally from those in rural regions where most utility-scale facilities are built. To ensure our findings are not driven by such idiosyncrasies, we re-run the baseline regression on several restricted subsamples. The results of these tests are shown in Table S13.

In Column (1), we addressed the potential influence of outliers by excluding the top and bottom 5% of observations for the dependent variable (county-level average Shannon Index). The resulting coefficient for the policy stringency index (PSI) was -0.0152 ($SE = 0.0039$, $p < 0.01$), confirming a robust negative impact. Similarly, in Column (2), removing the top 5% of observations with the highest PSI yielded a consistent and significant coefficient of -0.0123 ($SE = 0.0038$, $p < 0.01$). These tests indicate that our primary findings are not skewed by extreme values at either tail of the variable distributions.

Our study tested whether the observed biodiversity declines were driven by specific administrative regions by excluding all provincial capitals and municipalities directly under the central government. These major urban centers typically exhibit divergent economic structures, land constraints, and policy priorities compared to standard counties. After their removal, the estimated effect of PSI in Column (3) remained significantly negative ($\beta = -0.0099$, $SE = 0.0041$, $p < 0.05$). Collectively, these tests confirm that our central conclusion is robust and not an artifact of specific sample characteristics or outliers.

Robustness check-Excluding county-month observations below minimum birdwatching record thresholds

Unreliable diversity metrics derived from low observer effort might bias the observed relationship between policy implementation and bird diversity. To address this concern and ensure our findings are not driven by biases associated with low observer effort, we re-estimated our core model on restricted subsamples, excluding county-month observations with sparse birdwatching records. The outcomes of these robustness checks are presented in Table S14.

In Column (1), we addressed the potential influence of limited observer effort by excluding observations falling below the bottom 20% threshold of total birdwatching records per county-month. The resulting coefficient for the primary variable of interest, the policy stringency index (PSI), was -0.0079 ($SE = 0.0047$, $p < 0.10$), confirming a robust negative impact. Similarly, we sequentially excluded observations below the bottom 10%, 5%, and 3% thresholds in Columns (2), (3), and (4). Removing the bottom 10% of observations yielded a significant coefficient of -0.0079 ($SE = 0.0039$, $p < 0.05$). Further restricting the sample by excluding the bottom 5% and 3% produced consistently significant coefficients of -0.0081 ($SE = 0.0034$, $p < 0.05$) and -0.0081 ($SE = 0.0032$, $p < 0.05$), respectively.

Across all four exclusion thresholds, the estimated effect of PSI remained significantly negative, with the magnitude staying stable. Collectively, these tests demonstrate that our core results hold firm even when the sample is restricted to areas with high observer effort, thereby alleviating concerns regarding the reliability of the bird diversity metrics in our original analysis.

Robustness check-Eliminating confounding environmental policies

A critical challenge in establishing a causal link is the potential confounding influence of other environmental and energy policies implemented concurrently. During our study period, China launched several major macro-level initiatives, including the Central Environmental Inspection (CEI) campaign, targeted closures of coal-fired power plants, wind power expansion, and the establishment of National Green Finance Reform and Innovation Pilot Zones. If the spatial and temporal rollout of these policies correlates with photovoltaic policy stringency and independently affects bird diversity, for instance, by mitigating pollution or altering land use, omitting them could bias our estimates.

To address this concern, we explicitly control for these potential confounders by introducing specific policy indicator variables into our baseline specification. We evaluated the effects of these policies both individually and simultaneously to isolate the specific impact of PV policies from broader national environmental campaigns (Table S15). The results demonstrate the robust nature of our primary findings. When controlling for all four concurrent policies simultaneously, the coefficient for the policy stringency index (PSI) in Column (5) remains significant at -0.0127 ($SE = 0.0037$, $p < 0.01$). This estimate is stable across all specifications, showing virtually no deviation when the confounding policies are introduced individually in Columns (1)-(4). While the CEI policy exhibits a marginally significant positive association with bird diversity ($\beta = 0.0248$, $SE = 0.0128$, $p < 0.10$), suggesting localized ecological benefits from stringent environmental enforcement, the inclusion of this and other policy variables does not attenuate the estimated impact of PV policies. This comprehensive test provides evidence that the observed negative relationship between photovoltaic policy stringency and bird diversity is directly attributable to the PV initiatives themselves, rather than being a spurious artifact driven by the confounding influence of broader concurrent environmental or energy transitions.

Robustness check-Other robustness checks

Table S16 presents a statistical analysis designed to validate the relationship between the policy stringency index (PSI) and bird diversity (ShannonBD). The table compares three different regression models to ensure the study's main findings remain consistent under varying conditions. The dependent variable in all three columns is ShannonBD, while the key independent variable is PSI. The table also lists numerous control variables, including environmental factors like temperature and wind, demographic factors like population, and land-use metrics such as water, green space, farmland, and grassland.

The three columns represent specific modifications to the base model. Column (1) introduces PM_{2.5} as an added control variable to account for the impact of air pollution. Column (2) tests for spatial biases by excluding counties immediately adjacent to those with high policy stringency. Column (3) examines seasonal sensitivity by restricting the dataset to the peak bird breeding season, defined as May through July.

The most notable result from the table is the consistency of the policy stringency coefficient. Across all three specifications, PSI shows a statistically significant negative correlation with ShannonBD. In the first and second columns, the coefficients are negative 0.0125 (SE = 0.0037, $p < 0.01$) and negative 0.0132 (SE = 0.0049, $p < 0.01$), respectively. The third column, focusing on the breeding season, yields a coefficient of negative 0.0148 (SE = 0.0057, $p < 0.05$). This suggests that higher policy stringency is associated with a decrease in bird diversity, regardless of whether the model controls for air pollution, spatial spillover, or seasonal variations.

Supplementary text-Specific characteristics of the protected species and PV expansion

Our findings suggest that a significant number of species included in the *List of State Key Protected Wild Animals (2021)* are vulnerable to the expansion of photovoltaic (PV) infrastructure. Species listed in the 2021 protection catalog are disproportionately affected due to three primary mechanisms: visual misperception (the lake effect), dependence on open habitats, and morphological constraints regarding take-off mechanics.

1. Visual misperception and the lake effect: This phenomenon is driven by polarized light pollution. The dark, smooth surfaces of photovoltaic panels reflect horizontally polarized light that mimics the optical signature of water bodies. Migratory waterbirds, relying on visual cues for navigation, may misinterpret these solar arrays as aquatic habitats. This maladaptive behavioral response can lead to collision mortality (impact trauma from high-speed dives) or the expenditure of critical energy reserves during failed landing attempts.

2. Dependence on open habitats and foraging grounds: Open-country species require extensive, unobstructed landscapes for hunting and breeding. The installation of PV infrastructure results in habitat fragmentation and loss of foraging efficiency.

3. Morphological constraints and the grounding effect: This vulnerability is biomechanical. Many large heavy-bodied waterbirds lack the ability to take off vertically; they require a long, unobstructed "taxiing" distance across a water surface to generate sufficient lift. If these birds are lured into landing on a solar array due to the Lake Effect, they may become "grounded" or stranded. The physical constraints of the solar panels and the lack of a water surface prevent them from achieving the necessary speed for take-off, leaving them vulnerable to dehydration, starvation, or predation.

Supplementary text-Evaluate the relationship between PSI and ShannonBD based on a one-standard-deviation change

In terms of economic significance, when we evaluate the relationship between photovoltaic policy stringency index (PSI) and bird diversity (ShannonBD) based on a one-standard-deviation change, the effect is calculated as:

(Estimated coefficient /mean value of ShannonBD) $\times$ standard deviation of PSI $\times$ 100%.

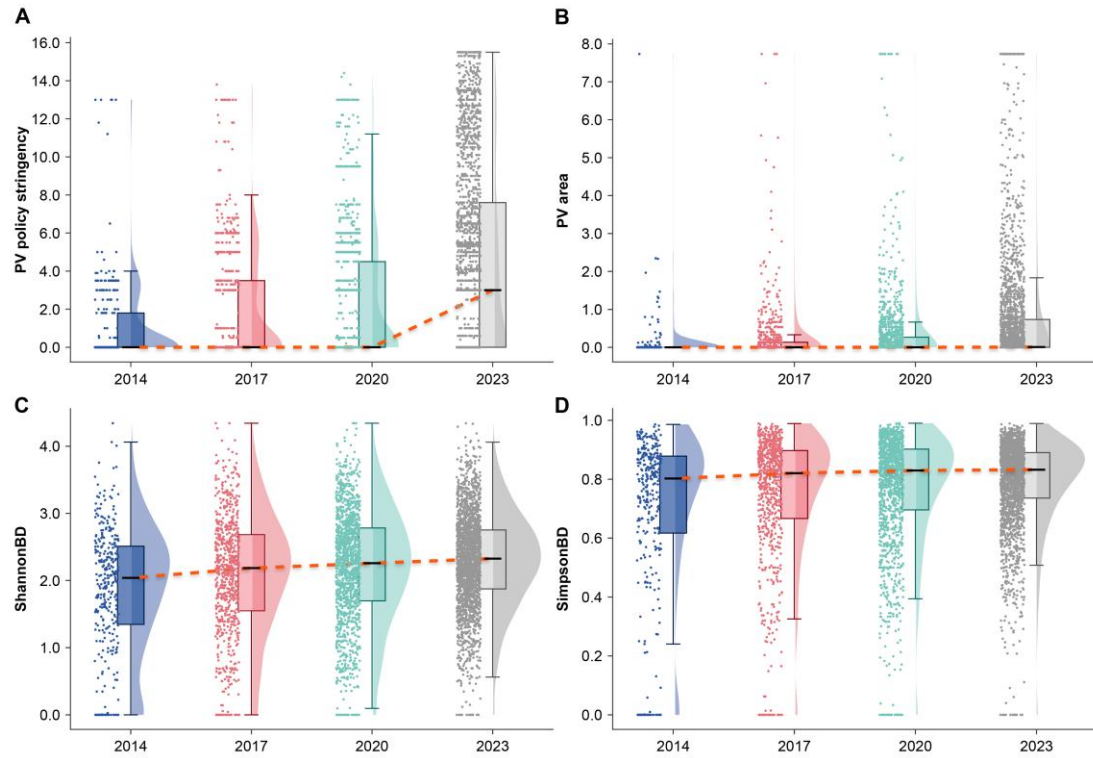


Fig. S1. The temporal evolution of China's solar boom and its ecological footprint. The figure presents rain cloud plots illustrating the distributions and trends of key variables across four representative years (2014, 2017, 2020, and 2023). Each panel combines box plots, density curves, and jittered data points to depict the full distributional characteristics, with the orange dashed line indicating the overall temporal trend. **(A)** Photovoltaic policy stringency index (PV policy stringency), measured as the weighted sum of policies at the provincial, municipal, and county levels. **(B)** An alternative measure of photovoltaic policy stringency (PV area), measured as the total area of centralized photovoltaic power stations. **(C)** Bird diversity (ShannonBD), measured as the negative sum of the proportion of each bird species multiplied by its natural logarithm. **(D)** An alternative measure of bird diversity (SimpsonBD), calculated as one minus the sum of the squared proportions of each bird species in the community.

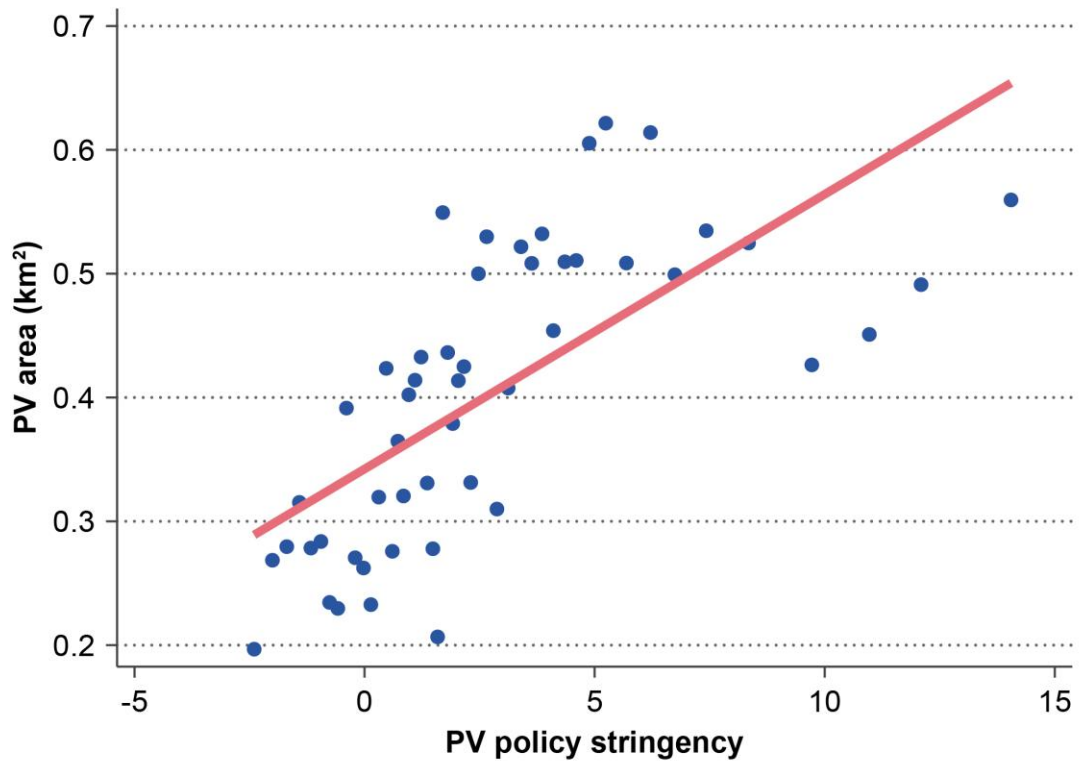


Fig. S2. The relationship between PV policy stringency and the physical footprint of solar infrastructure. This figure presents a binned scatter plot depicting the correlation between PV policy stringency (the weighted sum of policies at the provincial, municipal, and county levels) and PV area (the total area of centralized photovoltaic power stations, in km²). Each dot represents the mean value of a bin, and the red line indicates the linear fit. The positive slope suggests a statistically significant association between the two variables, confirming that regions with stronger PV policy stringency tend to exhibit larger physical deployment of solar infrastructure. The positive slope ($\beta = 0.0222$, $SE = 0.0026$, $p < 0.01$, $R^2=0.1124$) suggests a statistically significant association between the two variables. This strong correspondence validates the use of Area as an alternative measure of PV policy stringency in robustness analyses.

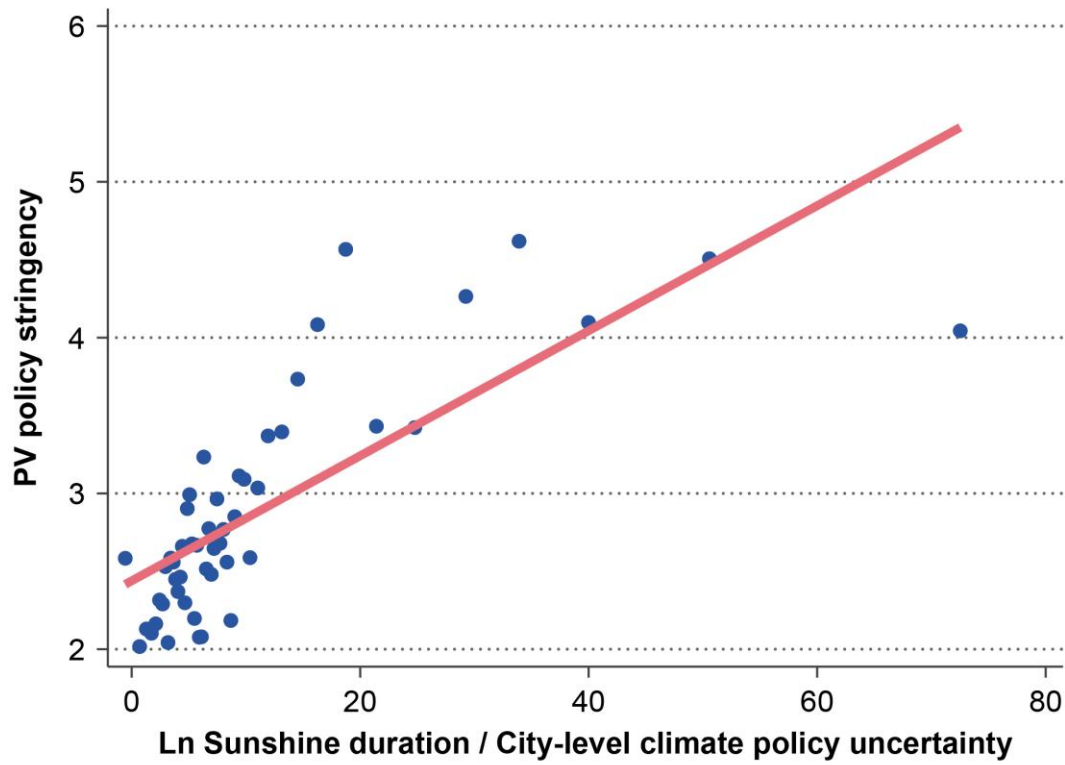


Fig. S3. Relationship between the instrumental variable and PV policy stringency. This figure presents a binned scatter plot illustrating the relationship between the instrumental variable and the endogenous variable (PV policy stringency) in the two-stage least squares (2SLS) estimation. The horizontal axis represents $\text{Ln}(\text{Sunshine}) / \text{city CPU}$, defined as the natural logarithm of the average sunshine duration from 1984 to 2013 divided by the city-level climate policy uncertainty (CPU) index. The city-level climate policy uncertainty index is sourced from Ma et al. (66). The vertical axis represents the photovoltaic policy stringency index, measured as the weighted sum of policies at the provincial, municipal, and county levels. Each dot represents the conditional mean of a bin, and the red line indicates the linear fit. The positive slope ($\beta = 0.0401$, $\text{SE} = 0.0028$, $p < 0.01$, $R^2 = 0.1844$) confirms that regions with greater sunshine endowment and less local climate policy uncertainty tend to exhibit stronger photovoltaic policy stringency.

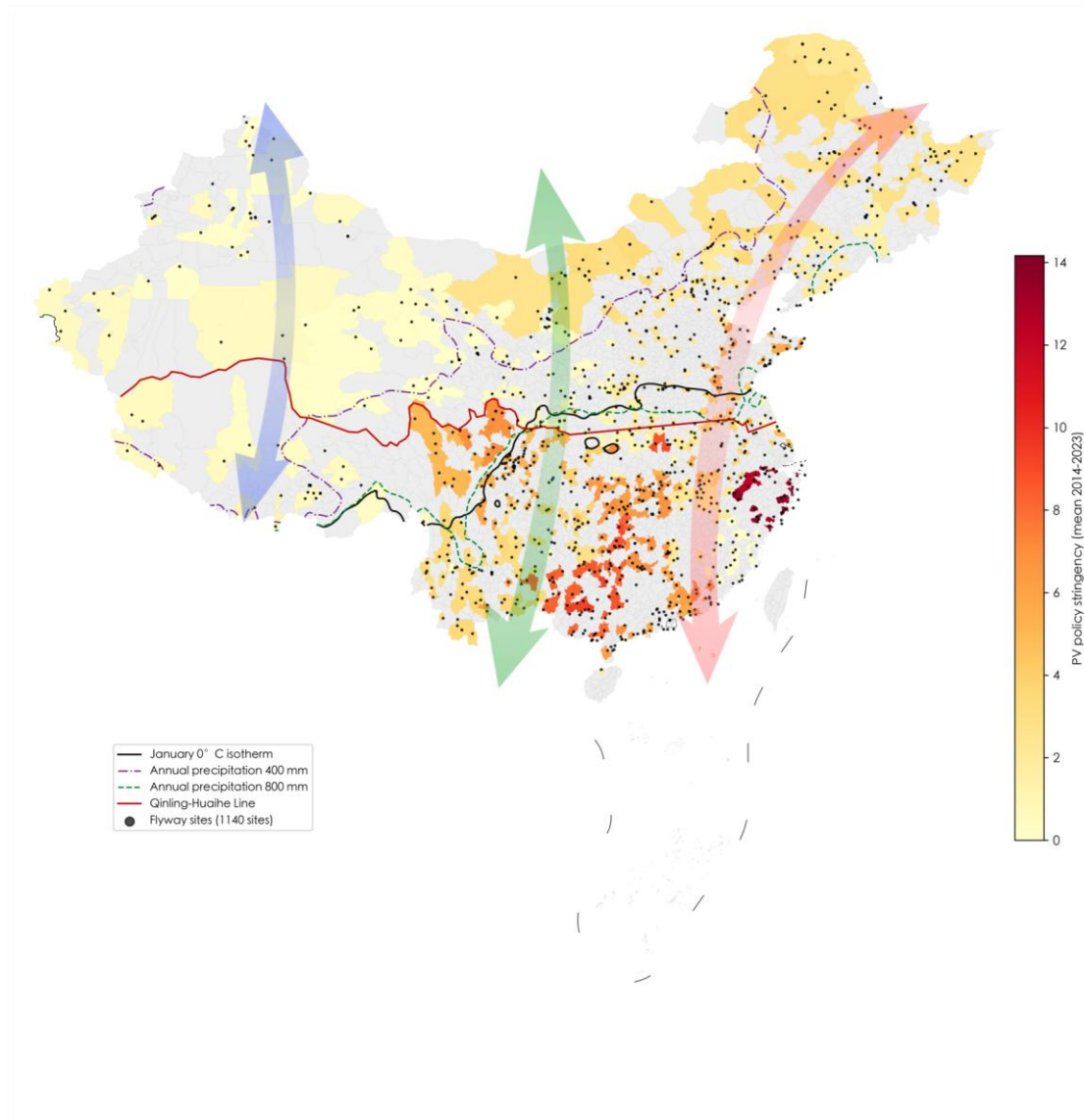


Fig. S4. Spatial distribution of county-level photovoltaic policy stringency and migratory-bird flyway stopover sites in China. Yellow and red shading indicate low and high PSI, respectively. Black dots represent stopover sites, while red, green, and blue arrows delineate major migratory flyways in eastern, central, and western China. Climatological and geographical boundaries include the January 0°C isotherm (solid black line), the 400-mm and 800-mm annual precipitation isohyets (purple and green dashed lines, respectively), and the Qinling – Huaihe Line (solid red line).

Table S1. Variable definitions. This table provides the definitions, measurements, and units of all variables used in the analysis. Variables with "—" in the Unit column are dimensionless indices, ratios, logarithmic transformations, or binary indicators.

Variable	Variable definition	Variable measurement	Unit
ShannonBD	Bird diversity	The negative sum of the proportion of each bird species multiplied by its natural logarithm	—
PSI	Photovoltaic policy stringency index	The weighted sum of policies at the provincial, municipal, and county levels	—
Temp	Annual average temperature	Annual average temperature	°C
Wind	Annual average wind speed	Annual average wind speed	m/s
Pop	Population density	The logarithm of the ratio of total population to regional area	—
Duration	Bird watching time per capita	The logarithm of the ratio of birdwatching duration to the number of birdwatchers	—
Carbon	Carbon emissions	The logarithm of the carbon emissions	—
Water	The proportion of water area	The area of water divided by the total area	—
Green	The proportion of forests and shrub area	The sum of the area of forests and shrubs divided by the total area	—
Farm	The proportion of farmland area	The area of farmland divided by the total area	—
Grass	The proportion of grassland area	The area of the grassland divided by the total area	—
Ln(Sunshine) / city CPU	Instrumental Variable	The natural logarithm of average sunshine duration from 1984 to 2013 divided by the city-level climate policy uncertainty index. The city-level climate policy uncertainty index is sourced from Ma et al. (66).	—
SimpsonBD	An alternative measure of bird diversity	One minus the sum of the squared proportions of each bird species in the community	—
Area	An alternative measure of photovoltaic policy stringency	The area of centralized photovoltaic power stations, calculated as (the installed capacity of centralized photovoltaic power stations × 1000) / (0.15 × 1,000,000)	km ²
CEI	Central environmental inspection policy	Equals 1 if an inspection team was stationed in the area in that year, and 0	—

		otherwise	
Coal	Coal power plant closure policy	Equals 1 if a coal-fired power plant was closed in the area in that year, and 0 otherwise	—
Windexp	Wind power expansion policy	Equals 1 if new wind power generation facilities were constructed in the area in that year, and 0 otherwise	—
Greenfin	Green finance pilot policy	Equals 1 if the region was designated as a National Green Finance Reform and Innovation Pilot Zone in the current year, and 0 otherwise	—
PM _{2.5}	PM _{2.5} concentration	The logarithm of PM _{2.5} concentration	—
NDVI	The Normalized Difference Vegetation Index	The difference between near-infrared and red band reflectance, divided by their sum, and scaled by a factor of 100	—
Nightlight	The Nighttime Light Index	Nightlight is computed from raster data using spatial statistics, with core indicators including total light value, average light intensity, and a composite light index	—
LAI	The Leaf Area Index	The total leaf area per unit of ground area	—
Sandy-Gobi	Sandy-Gobi dummy variable	Equals 0 if both sandy land and gobi are 0, and 1 otherwise	—
Species richness	Species richness	The total number of species observed within the survey area	—
Species evenness	Species evenness	The ratio of ShannonBD to the logarithm of species richness	—
Agricultural crop yield	Agricultural crop yield	The one-period lead of the logarithm of agricultural crop yield	—

Table S2. Data sources. This table summarizes the datasets and corresponding URLs for all variables used in the analysis.

Variable	Dataset	URL
ShannonBD	China Birdreport Online Database	www.birdreport.cn
PSI	The official government websites at provincial, municipal, and county levels	—
Temp	ERA5-Land hourly data from 1950 to present	https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview
Wind	ERA5-Land hourly data from 1950 to present	https://cds.climate.copernicus.eu/datasets/reanalysis-era5-land?tab=overview
Pop	National Bureau of Statistics	https://data.stats.gov.cn
Duration	China Birdreport Online Database	www.birdreport.cn
Carbon	Emissions Database for Global Atmospheric Research (EDGAR)	https://edgar.jrc.ec.europa.eu/
Water	The 30m resolution China land cover dataset	https://zenodo.org/records/12779975
Green	The 30m resolution China land cover dataset	https://zenodo.org/records/12779975
Farm	The 30m resolution China land cover dataset	https://zenodo.org/records/12779975
Grass	The 30m resolution China land cover dataset	https://zenodo.org/records/12779975
Ln(Sunshine) / city CPU	China Meteorological Data Service Centre; Ma et al. (66)	http://data.cma.cn ; Ma, Y.-R., Liu, Z., Ma, D. et al. A news-based climate policy uncertainty index for China. <i>Sci Data</i> 10, 881 (2023). https://doi.org/10.1038/s41597-023-02817-5 .
SimpsonBD	China Birdreport Online Database	www.birdreport.cn
Area	Based on statistics of enterprises that have obtained power generation business licenses from the National Energy Administration	https://www.powerdatahub.com/
CEI	Ministry of Ecology and Environment of the People's Republic of China	https://www.mee.gov.cn/ywgz/zysthjbhdc/
Coal	The Global Wind Power Tracker (GWPT)	https://globalenergymonitor.org/projects/global-wind-power-tracker/
Windexp	The Global Wind Power Tracker (GWPT)	https://globalenergymonitor.org/projects/global-wind-power-tracker/
Greenfin	International Institute of Green Finance, Central University of Finance and Economics	https://iigf.cufe.edu.cn/info/1012/9474.htm
PM _{2.5}	Satellite-derived PM _{2.5}	https://sites.wustl.edu/acag/surface-pm2-5/

NDVI	MOD13A3	https://search.earthdata.nasa.gov/search
Nightlight	An improved time-series DMSP-OLS-like data (1992-2019) in China by integrating DMSP-OLS and SNPP-VIIRS	https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/GIYGJU
LAI	MOD15A2H	https://search.earthdata.nasa.gov/search
Sandy-Gobi	China Multi-period Land Use Remote Sensing Monitoring Dataset (CNLUCC)	https://www.resdc.cn/DOI/DOI.aspx?DOIID=54
Species richness	China Birdreport Online Database	www.birdreport.cn
Species evenness	China Birdreport Online Database	www.birdreport.cn
Agricultural Crop yield	China Stock Market & Accounting Research Database (CSMAR)	https://data.csmar.com

Table S3. Scoring rubric for quantifying policy stringency. This table presents the scoring criteria used to assign numerical values to different types of policy instruments related to photovoltaic development. Scores range from 1 to 9, reflecting an ascending order of legal authority and binding force. These scores are used to construct the key independent variable PSI (the weighted sum of policies at the provincial, municipal, and county levels), enabling a standardized quantification of policy stringency that accounts for variation in the binding force of different policy instruments across administrative levels.

Policy Instrument Type	Scoring Criteria	Score
Policy Outline	Articulate broad principles and general guidance. Characterized by a low degree of compulsion and lack specific operational directives.	1
Strategic Plan	Establish long-term vision goals. While directive in intent, it serves primarily as a high-level strategic framework.	2
Key Work Priorities	Identify key areas of work and priority tasks. Carry a moderate level of compulsion, often requiring focus efforts and report on progress.	3
Guiding Opinion	Function as an advisory document intended to guide action rather than mandate it. It is suggestive and is not strongly binding.	4
Implementation Scheme	Provide detailed steps, programs, or projects for execution. Carries a moderate binding force to implement specific activities.	5
Official Notice	Formally communicate official requirements or directives to subordinate entities, with clear compulsory force.	6
Administrative Measures	Establish specific methods, procedures, or standards to manage a particular issue. It requires strict adherence to the prescribed protocols.	7
Regulation	Stipulate formal and detailed rules that govern specific activities or entities. Highly compulsory and legally enforceable.	8
Ordinance	A formal, law-like instrument possessing the highest degree of legal and administrative authority.	9

Table S4. Summary statistics for key variables. Panel A shows the descriptive statistics for the key variables covering the period from 2014 to 2023. ShannonBD refers to the bird diversity in the sample area (the negative sum of the proportion of each bird species multiplied by its natural logarithm). SimpsonBD refers to an alternative measure of bird diversity in the sample area (one minus the sum of the squared proportions of each bird species in the community). PSI refers to the stringency of photovoltaic policies in the sample area (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). City CPU (the city-level climate policy uncertainty index). Ln(Sunshine) (the natural logarithm of the average sunshine duration for each county during the period from 1984 to 2013). Panel B shows the correlation matrix of the main variables in this study. Each cell shows the Pearson correlation coefficient between the two variables.

Panel A: Descriptive statistics					
	Obs	Mean	S.D.	Min	Max
ShannonBD	46,528	2.4458	0.9814	0.0000	4.3425
SimpsonBD	46,528	0.8067	0.2202	0.0000	0.9833
PSI	46,528	2.6623	4.1159	0.0000	15.5000
Temp	46,528	14.8332	5.9125	-3.6423	24.5156
Wind	46,528	2.2355	0.8414	0.8092	4.8008
Pop	46,528	5.5680	1.8172	0.2308	9.5080
Duration	46,528	1.5019	1.1071	0.0000	5.8903
Carbon	46,528	14.5589	1.3927	11.3532	17.4598
Water	46,528	0.0358	0.0542	0.0001	0.2837
Green	46,528	0.3318	0.3197	0.0000	0.9377
Farm	46,528	0.3634	0.2457	0.0000	0.8882
Grass	46,528	0.0893	0.2108	0.0000	0.9103
city CPU	38,241	1.2132	0.7406	0.0908	3.2237
Ln(Sunshine)	45,784	7.5535	0.2596	6.9013	8.0227
Ln(Sunshine) / city CPU	37,503	12.3912	14.5068	2.2081	87.8951

Panel B: Pearson correlation matrix

	ShannonBD	PSI	Temp	Wind	Pop	Duration	Carbon	Water	Green	Farm	Grass	city CPU	Ln(Sunshine)
ShannonBD	1.00												
PSI	0.02***	1.00											
Temp	0.16***	0.11***	1.00										
Wind	-0.10***	-0.17***	0.09***	1.00									
Pop	0.04***	-0.23***	0.50***	0.32***	1.00								
Duration	0.23***	-0.01	-0.01	-0.07***	-0.07***	1.00							
Carbon	-0.01***	-0.11***	0.12***	0.34***	0.30***	-0.05***	1.00						
Water	0.04***	-0.15***	0.23***	0.35***	0.28***	-0.02***	0.25***	1.00					
Green	0.11***	0.30***	0.17***	-0.52***	-0.40***	0.10***	-0.37***	-0.37***	1.00				
Farm	-0.06***	0.01	0.19***	0.30***	0.29***	-0.07***	0.34***	0.18***	-0.59***	1.00			
Grass	-0.10***	-0.13***	-0.66***	-0.05***	-0.42***	-0.01***	-0.10***	-0.18***	-0.22***	-0.32***	1.00		
city CPU	-0.00	-0.19***	0.01**	0.02***	0.12***	0.07***	0.02***	0.12***	-0.07***	-0.05***	0.03***	1.00	
Ln(Sunshine)	-0.16***	-0.23***	-0.54***	0.35***	-0.22***	-0.04***	0.17***	0.02***	-0.33***	-0.08***	0.48***	-0.04***	1.00

Table S5. Descriptive statistics of bird observation records and total abundance categorized by heterogeneous functional groups. The table summarizes the sample sizes for each functional group used in the heterogeneity analysis. The categories include endemism status (Endemic vs. Non-endemic), migratory status (Migratory vs. Resident), conservation status (Protected vs. Non-protected), nesting guilds (Land-water vs. Vegetation), dietary guilds (Carnivorous and omnivorous vs. Herbivores), and flocking behavior (Small-flock vs. Large-flock). Observation frequency refers to the number of occurrence records, while Total individuals represent the cumulative count of birds. The Unknown category indicates records where specific functional trait data were unavailable.

Heterogeneous group	Category	Observation frequency	Total individuals
Endemism status	Endemic	236,782	969,241
	Non-endemic	6,592,654	3,546,563,149
Migratory status	Migratory	1,864,248	29,476,950
	Resident	4,965,188	3,518,055,440
Conservation status	Protected	652,836	5,838,311
	Non-protected	6,176,600	3,541,694,079
Nesting guilds	Land-water	2,544,745	37,260,262
	Vegetation	4,196,522	3,508,846,148
	Unknown	88,169	1,425,980
Dietary guilds	Carnivorous and omnivorous	6,753,373	3,547,246,986
	Herbivores	63,441	265,131
	Unknown	12,622	20,273
	Small-flock	5,886,204	3,528,627,106
Flocking behavior	Large-flock	818,080	18,658,528
	Unknown	125,152	246,756

Table S6. Pearson correlation coefficients between PSI at year t and Solar area at different time lags. PSI at year t is correlated with solar area at year $t + k$, where $k \in \{-3, -2, -1, 0, +1, +2, +3\}$. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

Lag (years)	$t-3$	$t-2$	$t-1$	t	$t+1$	$t+2$	$t+3$
Pearson r	0.0287***	0.0406***	0.0413***	0.0414***	0.0445***	0.0527***	0.0500***

Table S7. Instrumental variable estimates. Column (1) shows the result of the first-stage regression. The dependent variable in Column (1) is PSI (the weighted sum of policies at the provincial, municipal, and county levels). The independent variable in Column (1) is Ln(Sunshine) / city CPU, defined as the natural logarithm of the average sunshine duration from 1984 to 2013 divided by the city-level climate policy uncertainty (CPU) index. The city-level climate policy uncertainty index is sourced from Ma et al. (66). Column (2) shows the result of the second-stage regression. The dependent variable in Column (2) is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable in Column (2) is PSI. K-P rk LM statistic shows the result of underidentification test. K-P rk Wald F statistic shows the result of weak identification test. Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1) PSI	(2) ShannonBD
Ln(Sunshine) / city CPU	0.0252*** (0.0046)	
PSI		-0.1264*** (0.0347)
Temp	-0.0623 (0.0826)	-0.0039 (0.0245)
Wind	0.6206*** (0.2106)	0.1698** (0.0847)
Pop	-0.2241 (0.3240)	-0.1335 (0.1372)
Duration	-0.0273*** (0.0096)	0.1554*** (0.0064)
Carbon	-0.1220 (0.2072)	-0.1384** (0.0546)
Water	14.1078*** (3.7060)	0.5986 (1.2926)
Green	6.5362 (4.1107)	2.1733** (1.0528)
Farm	4.3525 (2.7758)	0.8204 (0.8162)
Grass	0.9890 (4.4269)	1.1258 (1.3907)
Year-month FE	YES	YES
County FE	YES	YES
Observations	37503	37503
K-P rk LM statistic	30.780***	

K-P rk Wald F statistic	30.181
Cragg-Donald Wald F statistic	664.359

Table S8. Exogeneity test of the shift component (city CPU). This table tests whether the shift component of the instrument, the city-level climate policy uncertainty index (CPU), is correlated with observable ecological and socioeconomic characteristics. The dependent variable is city CPU, sourced from Ma et al. (66). Column (1) reports a cross-sectional regression of city-level mean CPU on city-level mean covariates. Column (2) reports a panel regression controlling for city and year fixed effects with robust standard errors clustered by city. The key result is Column (2): after absorbing city and year fixed effects (consistent with the main IV specification), none of the covariates significantly predict the within-city time variation in CPU. Control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1) Cross-sectional OLS	(2) Panel with city and year FE
Temp	-0.0001 (0.0063)	0.0194 (0.0219)
Wind	-0.0309 (0.0338)	0.0189 (0.0832)
Pop	0.1119*** (0.0234)	-0.0432 (0.0461)
Duration	0.1814*** (0.0508)	0.0124 (0.0138)
Carbon	-0.0334 (0.0248)	-0.0217 (0.0219)
Water	0.9403* (0.4962)	1.6308** (0.7579)
Green	-0.6294*** (0.1579)	0.2787 (0.3788)
Farm	-0.9385*** (0.1627)	0.3704 (0.3444)
Grass	0.0503 (0.1938)	0.3696 (0.4614)
City FE		YES
Year FE		YES
Joint F-statistic		1.10
Joint F p-value		0.360
Observations	288	2294
R-squared	0.2377	0.5743
Within R-squared		0.0068

Table S9. Shift-share IV diagnostics: Overidentification and sensitivity analysis. This table reports the shift-share IV diagnostics following Goldsmith-Pinkham et al. (70). The instrument is $\text{Ln}(\text{Sunshine}) / \text{city CPU}$, where $\text{Ln}(\text{Sunshine})$ is the time-invariant share (natural logarithm of average sunshine duration for each county during the period from 1984 to 2013) and $1 / \text{city CPU}$ is the time-varying shift (inverse of city-level climate policy uncertainty index). Column (1) reports the overidentification test. We divide observations into terciles based on the CPU level and interact each tercile indicator with the original instrument $\text{Ln}(\text{Sunshine}) / \text{city CPU}$, yielding three instruments that exploit the same source of identifying variation — the within-city temporal fluctuation of climate policy uncertainty — but operate under different policy uncertainty regimes (low, medium, and high CPU). We estimate the overidentified model using `ivreghdfe` with cluster-robust standard errors, which automatically absorbs county and year-month fixed effects and reports the Hansen J statistic. The Hansen J statistic tests whether all instruments are jointly valid (H_0 : all instruments are valid; failure to reject supports instrument validity). Columns (2)–(5) present the sensitivity analysis: Column (2) uses the full sample; Column (3) excludes counties in the top and bottom 5% of the $\text{Ln}(\text{Sunshine})$ distribution; Column (4) excludes the top 10%; Column (5) excludes the bottom 10%. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The endogenous variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels), instrumented by $\text{Ln}(\text{Sunshine}) / \text{city CPU}$. Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and year-month fixed effects. The robust standard errors clustered by county are reported in parentheses. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	Over-identification	Full sample	Excl. extreme 5%	Excl. top 10%	Excl. bottom 10%
PSI	-0.1222*** (0.0336)	-0.1264*** (0.0347)	-0.1096*** (0.0309)	-0.1072*** (0.0292)	-0.1220*** (0.0361)
Temp	-0.0037 (0.0243)	-0.0039 (0.0245)	-0.0155 (0.0258)	-0.0059 (0.0253)	-0.0263 (0.0271)
Wind	0.1666** (0.0840)	0.1698** (0.0847)	0.1615* (0.0853)	0.1541* (0.0849)	0.1772** (0.0868)
Pop	-0.1326 (0.1368)	-0.1335 (0.1372)	-0.0717 (0.1483)	-0.1413 (0.1369)	-0.0970 (0.1494)
Duration	0.1555*** (0.0064)	0.1554*** (0.0064)	0.1567*** (0.0067)	0.1520*** (0.0066)	0.1562*** (0.0068)
Carbon	-0.1379** (0.0543)	-0.1384** (0.0546)	-0.1334** (0.0551)	-0.1271** (0.0561)	-0.1270** (0.0543)
Water	0.5347 (1.2834)	0.5986 (1.2926)	0.6113 (1.2589)	0.7639 (1.2730)	0.3158 (1.2681)
Green	2.1433** (1.0445)	2.1733** (1.0528)	2.1588** (1.0757)	2.0380* (1.0571)	1.8556* (1.1057)
Farm	0.7988 (0.8102)	0.8204 (0.8162)	0.8807 (0.8302)	0.7573 (0.8331)	0.8727 (0.8431)

Grass	1.1214 (1.3791)	1.1258 (1.3907)	1.9206 (1.3548)	1.1008 (1.6779)	1.0245 (1.3645)
Year-month FE	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES
Sample restriction	Over-identification	Full sample	Excl. extreme sunshine	Excl. top 10% sunshine	Excl. bottom 10% sunshine
Observations	37503	37503	34359	34941	34140
Hansen J statistic	0.6356	0.0000	0.0000	0.0000	0.0000
Hansen J p- value	0.7277				

Table S10. Share exogeneity and shift-based IV identification tests. This table reports the share exogeneity test and the shift-based IV identification results. Column (1) tests whether the time-invariant share component $\text{Ln}(\text{Sunshine})$ is correlated with observable county characteristics following Goldsmith-Pinkham et al. (70). The dependent variable in Column (1) is $\text{Ln}(\text{Sunshine})$, natural logarithm of average sunshine duration from 1984 to 2013. The data is collapsed to county-level means. If the share is exogenous, it should be uncorrelated with county-level covariates. Note that in the main IV specification, county fixed effects fully absorb the time-invariant cross-sectional variation in $\text{Ln}(\text{Sunshine})$, so any correlations observed here do not threaten identification. Columns (2)-(5) report the shift-based IV identification results following Borusyak et al. (69). Columns (2)-(3) use the original instrument $\text{Ln}(\text{Sunshine})/\text{CPU}$ but additionally control for $\text{Ln}(\text{Sunshine}) \times \text{Year-month FE}$, so that any time-varying effect of sunshine operating through channels other than climate policy is fully absorbed; identification relies exclusively on within-city temporal variation in CPU. Columns (4)-(5) use a pure shift instrument $1/\text{CPU}$, which does not involve the sunshine share at all. Columns (2) and (4) report first-stage regressions where the dependent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Columns (3) and (5) report second-stage regressions where the dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The slight variation in sample sizes across specifications reflects differences in the availability of the CPU index and sunshine data across county-year observations. Control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). Columns (2)-(5) include county and year-month fixed effects with robust standard errors clustered by county. Column (1) reports robust standard errors. ***, **, * denote significance at the 1%, 5%, and 10% levels, respectively.

	Balance test	Panel A: [$\text{Ln}(\text{Sunshine})$ / city CPU] and [$\text{Ln}(\text{Sunshine}) \times \text{Year-month FE}$]		Panel B: 1 / city CPU only	
	(1)	(2)	(3)	(4)	(5)
Share exogeneity		First stage	Second stage	First stage	Second stage
$\text{Ln}(\text{Sunshine})$ / city CPU		0.0291*** (0.0044)			
1 / city CPU				0.2054*** (0.0293)	
PSI			-0.1169*** (0.0288)		-0.0938*** (0.0291)
Temp	-0.0202*** (0.0008)	-0.1504* (0.0788)	-0.0069 (0.0241)	-0.0600 (0.0815)	-0.0029 (0.0233)
Wind	0.0926*** (0.0069)	0.9537*** (0.2317)	0.1845** (0.0922)	0.3476 (0.2328)	0.1289 (0.0807)
Pop	-0.0126*** (0.0031)	-0.2821 (0.3205)	-0.1192 (0.1343)	-0.2438 (0.3230)	-0.1253 (0.1341)
Duration	-0.0120* (0.0070)	-0.0263*** (0.0096)	0.1551*** (0.0065)	-0.0294*** (0.0096)	0.1552*** (0.0063)

Carbon	0.0238*** (0.0028)	0.0430 (0.1871)	-0.1069** (0.0527)	-0.0536 (0.2033)	-0.1143** (0.0517)
Water	-0.3906*** (0.0641)	17.8066*** (3.8446)	0.9475 (1.2933)	15.4788*** (3.7700)	0.3096 (1.2532)
Green	-0.3035*** (0.0215)	4.5472 (4.1782)	1.8035* (1.0191)	5.3258 (4.1523)	1.6576 (1.0177)
Farm	-0.2832*** (0.0228)	6.1941** (2.8800)	1.1296 (0.8167)	5.1342* (2.8275)	0.7884 (0.7949)
Grass	-0.0920*** (0.0201)	-5.9045 (4.2661)	0.1901 (1.2718)	1.5858 (4.4327)	1.1088 (1.3081)
Observations	2471	37503	37503	38099	38099
R-squared	0.5808	0.9067	0.0112	0.8973	0.0185
F-statistic	485.64	8.78	66.10	8.00	66.89
Year-month FE		YES	YES	YES	YES
County FE		YES	YES	YES	YES
Ln(Sunshine) x Year-month FE		YES	YES	NO	NO

Table S11. Propensity score matching. Column (1) presents the PSM results using 1:1 nearest neighbor matching with a caliper of 0.05. Column (2) reports the PSM results using 1:2 nearest neighbor matching with a caliper of 0.05. Column (3) shows the PSM results based on kernel matching. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
	ShannonBD	ShannonBD	ShannonBD
PSI	-0.0072* (0.0039)	-0.0083** (0.0038)	-0.0125*** (0.0037)
Temp	-0.0024 (0.0234)	0.0129 (0.0222)	-0.0074 (0.0181)
Wind	-0.0083 (0.0992)	-0.0420 (0.0907)	0.0429 (0.0736)
Pop	-0.1996 (0.2180)	-0.2396 (0.1971)	-0.0887 (0.1287)
Duration	0.1862*** (0.0075)	0.1822*** (0.0070)	0.1637*** (0.0057)
Carbon	-0.0865* (0.0522)	-0.0775 (0.0497)	-0.1119*** (0.0431)
Water	-1.3545 (1.2507)	-0.6969 (1.2236)	-1.2808 (1.1270)
Green	0.0350 (1.1828)	0.4795 (1.1420)	1.1436 (0.9520)
Farm	-0.3074 (0.9684)	0.0238 (0.8994)	0.0688 (0.7089)
Grass	0.5010 (1.4603)	0.6740 (1.3074)	1.2628 (0.9470)
Year-month FE	YES	YES	YES
County FE	YES	YES	YES
Observations	27202	30968	46361
R-squared	0.2912	0.2923	0.3000

Table S12. Robustness checks using alternative variable measures. This table presents the regression results with alternative measures of key variables. The dependent variable in Columns (1)-(7) is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). In Column (1), the core explanatory variable is replaced with Area (the total area of centralized photovoltaic power stations). Column (2) replaces it with policy stringency calculated using arithmetic progression scores of 1, 2, and 3 with hierarchical weights of 0.5, 0.3, and 0.2. Column (3) replaces it with geometric progression scores of 1, 2, and 4 with hierarchical weights of 0.5, 0.3, and 0.2. Column (4) replaces it with exponential progression scores of 2, 4, and 8 with hierarchical weights of 0.5, 0.3, and 0.2. Column (5) replaces it with arithmetic progression scores of 1, 2, and 3 with equal weights of 1. Column (6) replaces it with geometric progression scores of 1, 2, and 4 with equal weights of 1. Column (7) replaces it with exponential progression scores of 2, 4, and 8 with equal weights of 1. The dependent variable in Columns (8) is SimpsonBD (one minus the sum of the squared proportions of each bird species in the community). The independent variable in Column (8) is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Central Areas	PVWeighted Score 123	Weighted Score 124	Weighted Score 248	Unweighted Score 123	Unweighted Score 124	Unweighted Score 248	SimpsonBD
Area	-0.0387*** (0.0118)							
PSI		-0.0325*** (0.0090)	-0.0294*** (0.0080)	-0.0147*** (0.0040)	-0.0132*** (0.0037)	-0.0121*** (0.0034)	-0.0061*** (0.0017)	-0.0014* (0.0008)
Temp	-0.0071 (0.0180)	-0.0067 (0.0181)	-0.0065 (0.0181)	-0.0065 (0.0181)	-0.0063 (0.0181)	-0.0064 (0.0181)	-0.0064 (0.0181)	0.0009 (0.0042)
Wind	0.0403 (0.0737)	0.0434 (0.0736)	0.0433 (0.0736)	0.0433 (0.0736)	0.0423 (0.0735)	0.0426 (0.0736)	0.0426 (0.0736)	0.0013 (0.0171)
Pop	-0.0936 (0.1274)	-0.0889 (0.1286)	-0.0888 (0.1285)	-0.0888 (0.1285)	-0.0891 (0.1282)	-0.0888 (0.1283)	-0.0888 (0.1283)	-0.0103 (0.0279)
Duration	0.1641*** (0.0057)	0.1637*** (0.0057)	0.1637*** (0.0057)	0.1637*** (0.0057)	0.1638*** (0.0057)	0.1638*** (0.0057)	0.1638*** (0.0057)	0.0274*** (0.0013)
Carbon	-0.0994** (0.0430)	-0.1115*** (0.0430)	-0.1119*** (0.0430)	-0.1119*** (0.0430)	-0.1102** (0.0429)	-0.1109*** (0.0429)	-0.1109*** (0.0429)	-0.0128 (0.0103)
Water	-1.4108 (1.1226)	-1.2457 (1.1243)	-1.2539 (1.1240)	-1.2539 (1.1240)	-1.2416 (1.1226)	-1.2580 (1.1229)	-1.2580 (1.1229)	-0.1617 (0.2178)
Green	1.1905 (0.9473)	1.1525 (0.9517)	1.1448 (0.9516)	1.1448 (0.9516)	1.1608 (0.9506)	1.1508 (0.9509)	1.1508 (0.9509)	0.4314** (0.1879)
Farm	0.1600 (0.7051)	0.0720 (0.7098)	0.0584 (0.7095)	0.0584 (0.7095)	0.0962 (0.7084)	0.0795 (0.7084)	0.0795 (0.7084)	0.1913 (0.1468)
Grass	1.1292	1.2921	1.2934	1.2934	1.3268	1.3216	1.3216	0.0555

[illegible]

Table S13. Robustness checks using alternative sampling criteria. This table shows the results of robustness checks using alternative sampling criteria. Column (1) excludes the top and bottom 5% of the dependent variable based on county-level averages. Column (2) removes the top 5% of the independent variable based on county-level averages. Column (3) excludes provincial capital cities and municipalities directly under the central government. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1) ShannonBD	(2) ShannonBD	(3) ShannonBD
PSI	-0.0152*** (0.0039)	-0.0123*** (0.0038)	-0.0099** (0.0041)
Temp	-0.0153 (0.0189)	-0.0079 (0.0184)	-0.0061 (0.0217)
Wind	0.0863 (0.0771)	0.0350 (0.0760)	0.0048 (0.0935)
Pop	-0.0856 (0.1298)	-0.1333 (0.1241)	0.1300 (0.2259)
Duration	0.1617*** (0.0060)	0.1633*** (0.0059)	0.1871*** (0.0068)
Carbon	-0.1214** (0.0476)	-0.1164*** (0.0441)	-0.0941** (0.0454)
Water	-1.2160 (1.1652)	-1.3599 (1.1889)	-0.4262 (1.2813)
Green	1.4266 (0.9732)	1.1884 (1.0211)	2.6018** (1.0606)
Farm	0.1278 (0.7297)	-0.0967 (0.7488)	2.2190** (0.8938)
Grass	1.5146 (0.9476)	1.1768 (0.9665)	2.1981** (1.0626)
Year-month FE	YES	YES	YES
County FE	YES	YES	YES
Observations	41797	44048	33102
R-squared	0.2185	0.3045	0.2818

Table S14. Excluding county-month observations below minimum birdwatching record thresholds. Columns (1) to (4) report results after excluding county-month observations with low birdwatching activity to address potential measurement error from limited observer effort. Specifically, Column (1) excludes observations falling below the bottom 20% threshold of total birdwatching records per county-month. Columns (2), (3), and (4) exclude observations below the bottom 10%, 5%, and 3% thresholds, respectively. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), **Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers)**, Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)
	Trim bottom 20%	Trim bottom 10%	Trim bottom 5%	Trim bottom 3%
PSI	-0.0079* (0.0047)	-0.0079** (0.0039)	-0.0081** (0.0034)	-0.0081** (0.0032)
Temp	-0.0190 (0.0243)	-0.0328* (0.0199)	-0.0266 (0.0170)	-0.0275* (0.0154)
Wind	-0.0017 (0.1039)	0.0101 (0.0831)	0.0254 (0.0655)	0.0186 (0.0619)
Pop	-0.0715 (0.1682)	-0.0435 (0.1647)	-0.0527 (0.1302)	-0.0275 (0.1019)
Duration	0.0322*** (0.0059)	0.0524*** (0.0056)	0.0642*** (0.0047)	0.0693*** (0.0043)
Carbon	-0.0752 (0.0631)	-0.0429 (0.0467)	-0.0699* (0.0382)	-0.0622* (0.0367)
Water	1.1165 (1.7449)	0.2338 (1.2517)	0.4891 (1.0889)	0.2972 (0.9644)
Green	0.1207 (1.4519)	-0.0836 (0.8650)	0.2236 (0.7360)	0.1695 (0.7699)
Farm	-0.5212 (0.8543)	-0.5413 (0.5426)	-0.3796 (0.5100)	-0.2600 (0.5300)
Grass	4.0359** (1.6300)	2.3847** (1.0533)	1.5275 (1.0926)	1.4735 (1.0359)
Year-month FE	YES	YES	YES	YES
County FE	YES	YES	YES	YES
Observations	7954	15160	22314	26918
R-squared	0.4724	0.3929	0.3491	0.3296

Table S15. Ruling out confounding effects from related policies. This table presents the regression results after excluding the interference of related policies. Column (1) controls for the central environmental inspection policy indicator CEI (equals 1 if an inspection team was stationed in the area in that year, and 0 otherwise). Column (2) controls for the coal power plant closure policy indicator Coal (equals 1 if a coal-fired power plant was closed in the area in that year, and 0 otherwise). Column (3) controls for the wind power expansion policy indicator Windexp (equals 1 if new wind power generation facilities were constructed in the area in that year, and 0 otherwise), Column (4) controls for the green finance pilot policy indicator Greenfin (equals 1 if the region was designated as a National Green Finance Reform and Innovation Pilot Zone in the current year, and 0 otherwise), and Column (5) controls for all of the above policies simultaneously. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)
	ShannonBD	ShannonBD	ShannonBD	ShannonBD	ShannonBD
PSI	-0.0127*** (0.0037)	-0.0125*** (0.0037)	-0.0125*** (0.0037)	-0.0126*** (0.0037)	-0.0127*** (0.0037)
CEI	0.0248* (0.0128)				0.0251* (0.0128)
Coal		0.0790 (0.0705)			0.0811 (0.0699)
Windexp			0.0182 (0.0251)		0.0195 (0.0251)
Greenfin				-0.0194 (0.0559)	-0.0189 (0.0559)
Temp	-0.0081 (0.0181)	-0.0073 (0.0181)	-0.0073 (0.0181)	-0.0072 (0.0181)	-0.0080 (0.0181)
Wind	0.0596 (0.0746)	0.0453 (0.0735)	0.0438 (0.0736)	0.0431 (0.0737)	0.0616 (0.0745)
Pop	-0.0882 (0.1283)	-0.0901 (0.1284)	-0.0879 (0.1286)	-0.0874 (0.1286)	-0.0881 (0.1279)
Duration	0.1639*** (0.0057)	0.1638*** (0.0057)	0.1638*** (0.0057)	0.1638*** (0.0057)	0.1639*** (0.0057)
Carbon	-0.1123*** (0.0430)	-0.1143*** (0.0431)	-0.1125*** (0.0430)	-0.1116*** (0.0431)	-0.1157*** (0.0430)
Water	-1.2414 (1.1287)	-1.2628 (1.1256)	-1.2738 (1.1267)	-1.2903 (1.1291)	-1.2269 (1.1291)
Green	1.1860	1.1508	1.1601	1.1449	1.1877

	(0.9522)	(0.9530)	(0.9522)	(0.9532)	(0.9550)
Farm	0.0788	0.0873	0.0829	0.0525	0.1066
	(0.7102)	(0.7081)	(0.7078)	(0.7109)	(0.7107)
Grass	1.2790	1.2639	1.2697	1.2656	1.2862
	(0.9489)	(0.9472)	(0.9472)	(0.9469)	(0.9496)
Year-month FE	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES
Observations	46371	46371	46371	46371	46371
R-squared	0.2999	0.2999	0.2999	0.2999	0.3000

Table S16. Other robustness checks. Column (1) includes PM_{2.5} as an additional control variable. Column (2) excludes all counties immediately adjacent to high-PI counties. Column (3) conducts a seasonal robustness check by running the main model exclusively on data from the peak breeding season (e.g., May to July). The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1)	(2)	(3)
	ShannonBD	ShannonBD	ShannonBD
PSI	-0.0125*** (0.0037)	-0.0132*** (0.0049)	-0.0148** (0.0057)
Temp	-0.0043 (0.0182)	0.0098 (0.0251)	0.0082 (0.0306)
Wind	0.0293 (0.0748)	-0.0236 (0.1024)	0.1002 (0.1301)
Pop	-0.1163 (0.1270)	-0.2325 (0.1584)	-0.2334 (0.1468)
Duration	0.1637*** (0.0058)	0.1681*** (0.0079)	0.1733*** (0.0098)
Carbon	-0.1128*** (0.0431)	-0.0443 (0.0522)	-0.0939 (0.0639)
Water	-1.2122 (1.1193)	-1.2490 (1.5870)	-0.9405 (1.7554)
Green	1.0878 (0.9521)	0.1932 (1.2030)	1.7232 (1.4013)
Farm	0.0683 (0.7118)	0.1291 (0.9951)	0.4630 (1.0013)
Grass	1.2132 (0.9524)	0.5853 (1.3191)	1.2326 (2.0228)
PM _{2.5}	-0.0531 (0.0882)		
Year-month FE	YES	YES	YES
County FE	YES	YES	YES
Observations	45965	24342	11829
R-squared	0.3008	0.2828	0.3336

Table S17. Cross-sectional heterogeneity. Specification 1 divides the sample into Non-poverty counties vs. Poverty counties. Specification 2 divides the sample into Non-Three-North regions vs. Three-North regions. Specification 3 divides the sample into Non-sandy and gravel desert vs. Sandy and gravel desert. The bdiff row reports the results of the between-group coefficient difference test. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	Specification 1		Specification 2		Specification 3	
	(1)	(2)	(3)	(4)	(5)	(6)
	Non-poverty counties	Poverty counties	Non-Three-North regions	Three-North regions	Non-sandy and gravel desert	Sandy and gravel desert
PSI	-0.0132*** (0.0047)	0.0007 (0.0060)	-0.0196*** (0.0042)	0.0056 (0.0076)	-0.0165*** (0.0038)	0.0190 (0.0119)
Temp	0.0104 (0.0219)	-0.0015 (0.0336)	0.0088 (0.0225)	-0.0363 (0.0323)	-0.0140 (0.0203)	0.0853** (0.0430)
Wind	0.0636 (0.0790)	-0.1120 (0.2085)	0.0225 (0.0819)	0.2438 (0.2002)	0.0567 (0.0794)	0.0249 (0.1854)
Pop	-0.0898 (0.1282)	0.4532 (0.9063)	-0.1300 (0.1682)	0.0618 (0.1688)	-0.1185 (0.1568)	0.1882 (0.1614)
Duration	0.1477*** (0.0062)	0.2253*** (0.0127)	0.1572*** (0.0066)	0.1758*** (0.0117)	0.1563*** (0.0060)	0.2179*** (0.0164)
Carbon	-0.1064** (0.0449)	-0.2203 (0.1629)	-0.0923* (0.0559)	-0.1232* (0.0713)	-0.1139** (0.0528)	0.0186 (0.0603)
Water	-0.7030 (1.1621)	-5.5089 (4.9648)	-1.2933 (1.2203)	-0.2996 (3.4485)	-0.9192 (1.1677)	1.1535 (5.6236)
Green	1.6224 (1.0059)	1.3952 (3.1610)	0.3391 (1.0859)	4.0844* (2.2267)	1.1031 (1.0600)	-0.9902 (1.8543)
Farm	0.0121 (0.7362)	3.1569 (2.4737)	-0.5089 (0.8026)	2.0584 (1.4398)	0.0850 (0.8039)	1.2061 (1.4109)
Grass	2.1995** (1.0978)	1.7462 (2.1226)	-3.2858 (2.2156)	2.6775*** (1.0290)	0.3367 (1.2977)	2.1022** (0.9255)
bdiff	-0.014**		-0.025***		-0.035***	
Year-month FE	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES	YES
Observations	37665	8704	34829	11542	41355	5016
R-squared	0.2851	0.3626	0.2749	0.3304	0.3023	0.2776

Table S18. Bird species heterogeneity. Panel A presents the results classified by endemism status, migratory status, and conservation status in sequence. Columns (1) and (2) correspond to Endemic species vs. Non-endemic species. Columns (3) and (4) correspond to Migratory birds vs. Resident birds. Columns (5) and (6) correspond to Protected species vs. Non-protected species. Panel B presents the results classified by nesting guilds, dietary guilds, and flocking behavior in sequence. Columns (7) and (8) correspond to Land-water nesters vs. Vegetation nesters. Columns (9) and (10) correspond to Carnivorous and omnivorous species vs. Herbivores species. Columns (11) and (12) correspond to Small-flock species vs. Large-flock species. The dependent variable is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The independent variable is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

Panel A	Endemism status		Migratory status		Conservation status	
	(1)	(2)	(3)	(4)	(5)	(6)
	Endemic	Non-endemic	Migratory	Resident	Protected	Non-protected
PSI	0.0022 (0.0022)	-0.0127*** (0.0036)	-0.0116*** (0.0037)	-0.0128*** (0.0034)	-0.0139*** (0.0034)	-0.0138*** (0.0035)
Temp	-0.0136 (0.0117)	-0.0072 (0.0177)	0.0012 (0.0168)	-0.0255 (0.0165)	-0.0336** (0.0156)	-0.0137 (0.0175)
Wind	-0.0463 (0.0385)	0.0394 (0.0717)	0.0372 (0.0705)	0.0302 (0.0711)	0.0793 (0.0683)	0.0636 (0.0722)
Pop	0.0514 (0.0661)	-0.0621 (0.1248)	-0.0012 (0.1306)	-0.1113 (0.1194)	0.0139 (0.1165)	-0.0980 (0.1211)
Duration	0.0333*** (0.0030)	0.1604*** (0.0055)	0.1256*** (0.0050)	0.1436*** (0.0052)	0.1176*** (0.0046)	0.1503*** (0.0054)
Carbon	-0.0529* (0.0307)	-0.1271*** (0.0407)	-0.1057*** (0.0370)	-0.1356*** (0.0371)	-0.0708* (0.0409)	-0.1433*** (0.0412)
Water	0.2362 (0.5912)	-1.2173 (1.1115)	0.0243 (1.0717)	-1.7266 (1.0697)	-1.3131 (0.8118)	-1.1137 (1.1616)
Green	0.1231 (0.5586)	1.3025 (0.9281)	0.0559 (0.8823)	1.0553 (0.8886)	0.6721 (0.8433)	1.1165 (0.9199)
Farm	-0.9720*** (0.3645)	0.1987 (0.6934)	-0.3724 (0.6887)	-0.1555 (0.6371)	-0.2391 (0.5733)	-0.0929 (0.6836)
Grass	0.4227 (0.4183)	1.4391 (0.8996)	1.2435 (1.2256)	1.3759* (0.7498)	1.1624 (0.7139)	1.3432 (0.8200)
Year-month FE	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES	YES
Observations	24581	46217	39837	45469	33562	45666
R-squared	0.3910	0.3040	0.2927	0.3340	0.3343	0.2997

Panel B	Nesting guilds		Dietary guilds		Flocking behavior	
	(7)	(8)	(9)	(10)	(11)	(12)
	Land-water	Vegetation	Carnivorous and omnivorous	Herbivorous	Small-flock	Large-flock
PSI	-0.0124*** (0.0034)	-0.0118*** (0.0035)	-0.0130*** (0.0036)	-0.0028 (0.0029)	-0.0127*** (0.0035)	-0.0103*** (0.0027)
Temp	-0.0088 (0.0162)	-0.0183 (0.0166)	-0.0108 (0.0177)	-0.0116 (0.0112)	-0.0116 (0.0176)	0.0043 (0.0123)
Wind	0.0646 (0.0687)	0.0211 (0.0681)	0.0443 (0.0718)	-0.0503 (0.0556)	0.0534 (0.0730)	0.0421 (0.0500)
Pop	-0.0839 (0.1150)	-0.1474 (0.1093)	-0.0661 (0.1262)	0.1757*** (0.0552)	-0.0856 (0.1229)	0.0029 (0.0848)
Duration	0.1322*** (0.0047)	0.1421*** (0.0052)	0.1597*** (0.0056)	0.0321*** (0.0039)	0.1585*** (0.0056)	0.0702*** (0.0036)
Carbon	-0.1093*** (0.0343)	-0.0631* (0.0361)	-0.1295*** (0.0419)	0.0124 (0.0293)	-0.1345*** (0.0392)	-0.0554* (0.0289)
Water	-1.1150 (1.1326)	-0.8465 (1.0439)	-1.1985 (1.1272)	0.4869 (0.8207)	-1.8318 (1.1565)	0.3075 (0.6881)
Green	1.1195 (0.7648)	0.5767 (0.9057)	1.3827 (0.9246)	-0.4372 (0.5728)	1.0046 (0.8997)	1.0556 (0.6982)
Farm	0.0887 (0.6346)	-0.2578 (0.6155)	0.1847 (0.6901)	1.3181** (0.5573)	0.1060 (0.6737)	0.0357 (0.5077)
Grass	2.1397** (0.9690)	0.5913 (0.7539)	1.4419* (0.8740)	0.8001 (0.6459)	1.1138 (0.8592)	1.7194** (0.8764)
Year-month FE	YES	YES	YES	YES	YES	YES
County FE	YES	YES	YES	YES	YES	YES
Observations	43999	44779	46259	9765	46045	38658
R-squared	0.2940	0.3458	0.3048	0.4084	0.3149	0.3098

Table S19. Further discussion. Column (1) includes Lag one PSI (the one-period lagged term of PSI). The dependent variable in Columns (1) is ShannonBD (the negative sum of the proportion of each bird species multiplied by its natural logarithm). The dependent variable in Column (2) is the species richness (the total number of species observed within the survey area). The dependent variable in Column (3) is the species evenness (the ratio of ShannonBD to the logarithm of species richness). The dependent variable in Column (4) is the agricultural crop yield (the one-period lead of the logarithm of agricultural crop yield). The independent variable in Column (1) to (4) is PSI (the weighted sum of policies at the provincial, municipal, and county levels). Other control variables include Temp (annual average temperature), Wind (annual average wind speed), Pop (the logarithm of the ratio of total population to regional area), Duration (the logarithm of the ratio of birdwatching duration to the number of birdwatchers), Carbon (the logarithm of the carbon emissions), Water (the area of water divided by the total area), Green (the sum of the area of forests and shrubs divided by the total area), Farm (the area of farmland divided by the total area), Grass (the area of the grassland divided by the total area). All specifications include county and Year-month fixed effects. The robust standard errors clustered by the county are reported in parenthesis. ***, **, * denote the significance at the 1%, 5%, and 10% levels, respectively.

	(1) ShannonBD	(2) Species richness	(3) Species evenness	(4) Agricultural crop yield
PSI	-0.0105** (0.0044)	-0.9751*** (0.1828)	0.0018*** (0.0007)	-0.0288*** (0.0081)
Lag one PSI	-0.0063 (0.0065)			
ShannonBD				0.0031 (0.0046)
Temp	-0.0288 (0.0200)	-1.0239 (0.7252)	0.0019 (0.0032)	-0.0066 (0.0259)
Wind	0.0789 (0.0804)	0.8314 (2.8575)	0.0027 (0.0131)	0.9882*** (0.1694)
Pop	-0.0707 (0.1403)	-4.3662 (5.6968)	0.0034 (0.0210)	0.2691 (0.1959)
Duration	0.1581*** (0.0063)	5.1542*** (0.1960)	0.0012 (0.0008)	-0.0039 (0.0042)
Carbon	-0.1231** (0.0580)	-4.3869** (1.7215)	0.0037 (0.0078)	-0.0062 (0.0702)
Water	-1.8330 (1.1414)	-36.9894 (51.2908)	0.1413 (0.1784)	3.7060 (2.5581)
Green	1.0643 (1.0883)	31.7306 (43.7458)	0.1594 (0.1267)	-0.9286 (1.4255)
Farm	-0.5004 (0.7322)	25.6639 (34.2553)	0.1675 (0.1133)	0.5292 (1.1097)
Grass	0.6144 (1.1356)	107.4241*** (36.8349)	0.0239 (0.1561)	3.4269** (1.6962)
Year-month FE	YES	YES	YES	YES
County FE	YES	YES	YES	YES
Observations	38497	45606	43958	15934

R-squared	0.2983	0.4610	0.1918	0.9372
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